

1 Estimation of Liquidity Demand Shock

We calibrate our idiosyncratic liquidity demand shock θ_{it} following Telyukova (2013)'s regression specification:

$$\log(c_{it}^{liq}) = \beta \mathbf{X}_{it} + u_i + \varepsilon_{it} \quad (1)$$

Where $\mathbf{X}_{it}$ is a vector of household observables including a cubic age, education, marital status, race, annual earnings, income before tax, family size, homeownership status, as well as a set of year and month dummies. u_i is the household fixed effect. ε_{it} is the idiosyncratic component of liquidity demand, which we assume evolves following the process

$$\varepsilon_{it} = \alpha \varepsilon_{it-1} + \eta_{it} \quad (2)$$

We use α exclusively for the persistence of the empirical first-stage residual, and η_{it} for its innovation. This notation distinguishes α from the model shock-persistence parameter ρ introduced below.

Let the model liquidity-demand shock follow

$$\log(\theta_{it}) = \rho \log(\theta_{i,t-1}) + \xi_{it}, \quad \text{Var}(\xi_{it}) = \sigma_\theta^2. \quad (3)$$

The estimated $\text{sd}(\hat{\eta}_{it})$ is interpreted as the empirical counterpart of the unconditional volatility of $\log(\theta_{it})$, rather than the volatility of the model innovation ξ_{it} . Writing $s_\eta = \text{sd}(\hat{\eta}_{it})$ and imposing covariance stationarity gives

$$s_\eta^2 = \text{Var}(\log(\theta_{it})) = \frac{\sigma_\theta^2}{1 - \rho^2}, \quad (4)$$

so that $\sigma_\theta = s_\eta \sqrt{1 - \rho^2}$. The model persistence ρ is calibrated separately by matching an adjacent-month liquid-consumption correlation, as described in a later section.

The empirical measure c_{it}^{liq} is the aggregate monthly expenditure on liquid goods for household i . Telyukova (2013)'s definition of liquid goods include: food, alcohol, tobacco, rents, mortgages, utilities, household repairs, childcare expenses, other household operations, property taxes, insurance, public transportation, and health insurance. Two alternate definitions that exclude food ¹, and further excludes property taxes are included as sensitivity checks. Gift purchases are excluded and components are deflated by their respective CPI values before entering the regression. The estimation of the c_{it}^{liq} term from the CEX data is detailed in the following sections.

¹Telyukova (2013)'s definition of food includes food, alcohol, and tobacco.

1.1 Construction of Monthly Liquid Consumption c_{it}^{liq}

1.1.1 CEX Survey Catalog

The two components of the CEX survey we use for constructing monthly liquid consumption and the subsequent regression are the 2018 and 2019 MTBI and FMLI interview files. MTBI files contain monthly expenditure records at the household level. Expenditure categories are uniquely identified by Universal Classification codes (UCCs), and each row of the MTBI dataset contains one household interview for one reference month-year and one UCC expenditure category. The MTBI dataset also includes a **GIFT** flag which indicates whether the item was purchased for someone outside the household, which is the flag we use to identify and exclude gift purchases as in Telyukova (2013). FMLI files are quarterly panels of respondent household characteristics including demographics, income, quarterly aggregate expenditures, and household survey weights. Variables in $\mathbf{X}$ are sourced from the FMLI files.

When constructing c_{it}^{liq} , Telyukova’s liquid consumption groups are identified from the monthly MTBI expenditure records using UCCs. CPI deflation is applied at the individual record UCC level. Expenditure records are first grouped by UCCs into detailed categories, then detailed categories are grouped into CEX expenditure components, which are finally grouped into Telyukova liquid consumption groups.

Table 1 records the relationship between UCCs, CEX detailed categories, CEX components, and final Telyukova consumption groups for all liquid consumption categories used in c_{it}^{liq} . Multiple UCCs appear in the same row only when they share the same downstream mapping.

Table 1: CEX UCC hierarchy used to construct Telyukova liquid-consumption groups

CEX UCC	CEX detailed category	CEX component	Telyukova group
190904, 790220, 790230, 190901, 190902, 190903, 790410, 790430, 800700	food_total	food	food_alcohol_tobacco_proxy
200900, 790310, 790320, 790420	alcohol	alcohol	food_alcohol_tobacco_proxy
630110, 630210	tobacco	tobacco	food_alcohol_tobacco_proxy
210110, 230121, 230141, 230150, 240111, 240121, 240211, 240221, 240311, 240321, 320611, 320621, 320631, 350110, 790690, 990920	rent_excluding_as_pay	rents	rent
800710	rent_as_pay	rents	rent
220311, 220313, 220321, 880110	mortgage_interest	mortgages	mortgage
210901, 220121, 220901, 230112, 230113, 230114, 230115, 230122, 230142, 230151, 230901, 240112, 240122, 240212, 240213, 240222, 240312, 240322, 320612, 320622, 320632, 340911, 990930	owned_dwelling_repairs_insurance_other	owned_dwelling_repairs_insurance_other	owned_dwelling_repairs_insurance_other
260211, 260212, 260213, 260214, 260111, 260112, 260113, 260114, 250111, 250112, 250113, 250114, 250211, 250212, 250213, 250214, 250221, 250222, 250223, 250224, 250901, 250902, 250903, 250904, 270101, 270102, 270103, 270104, 270211, 270212, 270213, 270214, 270411, 270412, 270413, 270414, 270901, 270902, 270903, 270904	utilities	utilities	utilities

continued on next page

Table 1 – *continued from previous page*

CEX UCC	CEX detailed category	CEX component	Telyukova group
340310, 340410, 340420, 340520, 340530, 340903, 340906, 340910, 340914, 340915, 330511, 340510, 340620, 340630, 340901, 340907, 340908, 690113, 690114, 990900	household_operations_total	household_operations	household_operations
340211, 340212, 670310	childcare_within_household_operations	childcare	childcare
220211	property_taxes	property_taxes	property_taxes
530110, 530210, 530312, 530411, 530510, 530901, 530311, 530412, 530902	public_transportation	public_transportation	public_transportation
580111, 580112, 580113, 580114, 580311, 580312, 580901, 580903, 580904, 580905, 580906	health_insurance	health_insurance	health_insurance

1.2 Liquid Payment Instrument Shares

While the CEX contains the monthly household expenditures for the aforementioned liquid good categories, it does not detail whether or not these expenditures were indeed made through liquid payment instrument. For example, we might observe that some households purchased a liquid good, such as food, using a non-liquid payment instrument, such as a credit card. Thus, in order to obtain a more accurate estimate of liquid consumption, we estimate a “share of good purchased via liquid payment instruments” using the Fed Survey and Diary of Consumer Payment Choice (SPCDC or DCPC for short), and weigh down each individual Telyukova consumption category by its corresponding share. If there is no exact match between a Telyukova consumption category and the DCPC, we proxy the Telyukova consumption category using the reasonably most similar DCPC consumption category.

1.2.1 DCPC Survey Catalog

The 2019 DCPC is a three-day transaction diary covering consumer payments during October 2019. Each row in the transaction-level file represents a reported transaction or other financial activity and records the respondent, diary day, amount, payment instrument, and merchant category. For specific merchant categories, additional pay codes provided more detail on the specific purposes of the transaction. Transactions are grouped into DCPC proxy categories based on these identifier columns, and are also assigned a day-of-week weight to account for heterogeneity in weekly purchases. Shares by proxy category are then computed by dividing the total dollar amount of transactions made with liquid payment instruments with the total dollar amount of transactions made with all instruments. We take the definition of liquid instruments to be any payment made with cash, check, or debit, as well as payments whose instruments are identified as bank-account-number payments, online banking bill pay, and account-to-account transfers. Detailed DCPC proxy category construction rules are in Table 3.

After identifying the mapping between the DCPC consumption category and the Telyukova consumption category used in the regression, we use the computed shares to weigh down the corresponding liquid consumption amounts in the regression. Table 2 records the mappings between DCPC consumption categories and Telyukova consumption categories.

The following flow diagram summarizes the group construction logic that eventually joins at the Telyukova consumption group level, where the consumption share weighting is applied.

Table 2: Mapping of DCPC proxy categories to Telyukova consumption groups

DCPC proxy category	Telyukova group
food_grocery_proxy	food_alcohol_tobacco_proxy
food_restaurant_bar_proxy	food_alcohol_tobacco_proxy
food_fast_service_proxy	food_alcohol_tobacco_proxy
rent	rent
mortgage_payment	mortgage
utilities	utilities
communications_entertainment_proxy	utilities
household_operations_bill_proxy	household_operations
household_repairs	owned_dwelling_repairs_insurance _other
homeowners_renters_insurance	owned_dwelling_repairs_insurance _other
property_taxes	property_taxes
public_transport_tolls	public_transportation
taxi_air_delivery_proxy	public_transportation
health_insurance	health_insurance
childcare	childcare
cash_contributions	cash_contributions

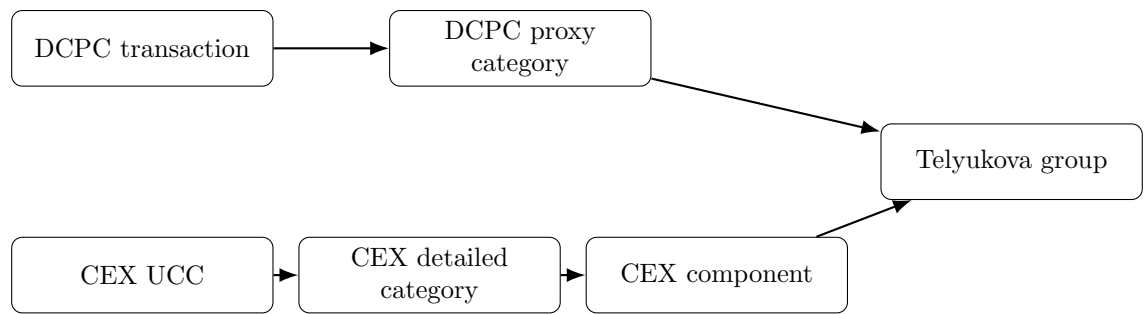


Table 3: Identification of DCPC proxy consumption categories

DCPC proxy category	Merchant code and meaning	Follow-up identification rule and meaning
<code>food_grocery_proxy</code>	merch=1: Grocery stores, convenience stores without gas stations, and pharmacies.	None. The merchant code alone identifies the proxy category.
<code>food_restaurant_bar_proxy</code>	merch=3: Sit-down restaurants and bars.	None. The merchant code alone identifies the proxy category.
<code>food_fast_service_proxy</code>	merch=4: Fast-food restaurants, coffee shops, cafeterias, and food trucks.	None. The merchant code alone identifies the proxy category.
<code>rent</code>	merch=14: Rent for apartments, homes, or other buildings, including payments to real-estate companies and property managers.	None. The merchant code alone identifies the category.
<code>mortgage_payment</code>	merch=15: Financial-services providers, including mortgage companies, banks, credit-card companies, and insurance companies.	pay010=2: Purpose was to make a loan payment; and pay011=1: the loan was a mortgage.
<code>utilities</code>	Either merch=8: Nongovernment utilities, including electricity, natural gas, water, sewer, trash, and heating oil; or merch=19: Government taxes or fees.	For merch=8 , no follow-up is required. For merch=19: pay040=1 means a government purchase of goods or services, and pay041=1 identifies electricity, water, or sewer.
<code>communications_entertainment_proxy</code>	merch=10: Telephone, internet, cable or satellite television, streaming services, and movie theaters.	None. The merchant code alone identifies the proxy category.
<code>household_operations_bill_proxy</code>	merch=6: General services; or merch=11: Building contractors, plumbers, electricians, and HVAC providers.	from_bill_section=1: The transaction originated in the bill-reminder section. These records are assigned to household operations before the general contractor rule is evaluated.
<code>household_repairs</code>	merch=11: Building contractors, plumbers, electricians, and HVAC providers.	No additional follow-up. This category contains the remaining merch=11 transactions after the bill-reminder household-operations rule has been applied.

DCPC proxy category	Merchant code and meaning	Follow-up identification rule and meaning
homeowners_renters_insurance	merch=15: Financial-services provider.	pay010=3: Purpose was to pay for insurance; and either pay016=1: homeowners insurance, or pay016=2: renters insurance.
property_taxes	merch=19: Government taxes or fees.	pay040=2: Purpose was to pay taxes; and pay042=4: the tax was a property tax.
public_transport_tolls	Either merch=21: Public transportation and tolls; or merch=19: Government taxes or fees.	For merch=21 , no follow-up is required. For merch=19: pay040=1 means a government purchase of goods or services, and either pay041=5 identifies tolls or pay041=7 identifies public transportation.
taxi_air_delivery_proxy	merch=9: Taxis, airplanes, and delivery services.	None. The merchant code alone identifies the proxy category.
health_insurance	One of three merchant codes: merch=15 , financial services; merch=18 , medical-care providers; or merch=19 , government.	Three alternative rules: (1) merch=15 , pay010=3 for insurance, and pay016=3 for health insurance; (2) merch=18 and pay030=4 , identifying the payee as an insurance company; (3) merch=19 , pay040=1 for government goods or services, and pay041=8 for out-of-pocket health insurance, including Medicare supplemental insurance.
childcare	Either merch=20: Schools, colleges, and childcare centers; or merch=19: Government taxes or fees.	For merch=20 , pay020=3 identifies childcare. For merch=19 , pay040=1 identifies government goods or services, and either pay041=3 identifies daycare or pay041=9 identifies childcare.
cash_contributions	merch=17: Charitable or religious donations.	Either pay050=1: donation; or pay050=2: offering, tithe, or money placed in a collection plate.

1.3 Data Innovation Standard Deviation $\text{sd}(\eta_{it})$

Table 4 reports $\hat{\alpha}$ and $\text{sd}(\hat{\eta}_{it})$ from equations (1) and (2) using monthly CEX data from 2018–2019. We include the two aforementioned alternate definitions of c_{it}^{liq} for sensitivity checks. Both stages and the reported innovation standard deviations are unweighted.

Table 4: Monthly empirical innovation-volatility estimates, 2018–2019

Liquid-consumption measure	$\hat{\alpha}$	$\text{sd}(\hat{\eta}_{it})$	$\text{sd}(\hat{\eta}_{it})$ (%)
Benchmark	0.141	0.271	27.14
Excluding food, alcohol, and tobacco	0.139	0.285	28.53
Excluding food, alcohol, tobacco, and property taxes	0.109	0.325	32.47

1.4 Adjacent-month Liquid Consumption Correlation for Moment Matching

The model persistence ρ is calibrated by matching the correlation between liquid consumption in consecutive months. Table 5 reports the 2019 correlations in levels and logs. These are pooled, unweighted correlations across strictly consecutive months for which both consumer-unit observations have positive liquid consumption.

Table 5: Adjacent-month liquid-consumption correlations, 2019

Liquid-consumption measure	Adjacent pairs	Households	$\text{Corr}(c_{it}^{liq}, c_{i,t-1}^{liq})$	$\text{Corr}(\log c_{it}^{liq}, \log c_{i,t-1}^{liq})$
Benchmark	51,095	11,432	0.589	0.852
Excluding food, alcohol, and tobacco	50,951	11,397	0.560	0.844
Excluding food, alcohol, tobacco, and property taxes	50,913	11,387	0.509	0.810